This procedure is applicable to below models:

V810 STD	✓	S1	✓	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	10 Apr 2015	First Release	CHIN CHEE YAN
01	11 Jan 2017	Content Revise	MUHAMMED SHAMEEM
02	13 Jan 2017	Template Changes & Content Updates	ONG KEAN SHENG
03	12 Sep 2017	Configuration Figure Revise	ONG KEAN SHENG

Content

A. Filter Height Sensor Checking

1. Prepare the 4XASS-357 Filter Height Jig as shown in the Figure 1

Figure 1 4XASS-357 Variable Mag Filter Height Jig

2. Check board existence in machine. Unload if any

3. Go to Services -> Subsystem Status & Control -> Panel Handling and click Initialize as shown in the Figure 2

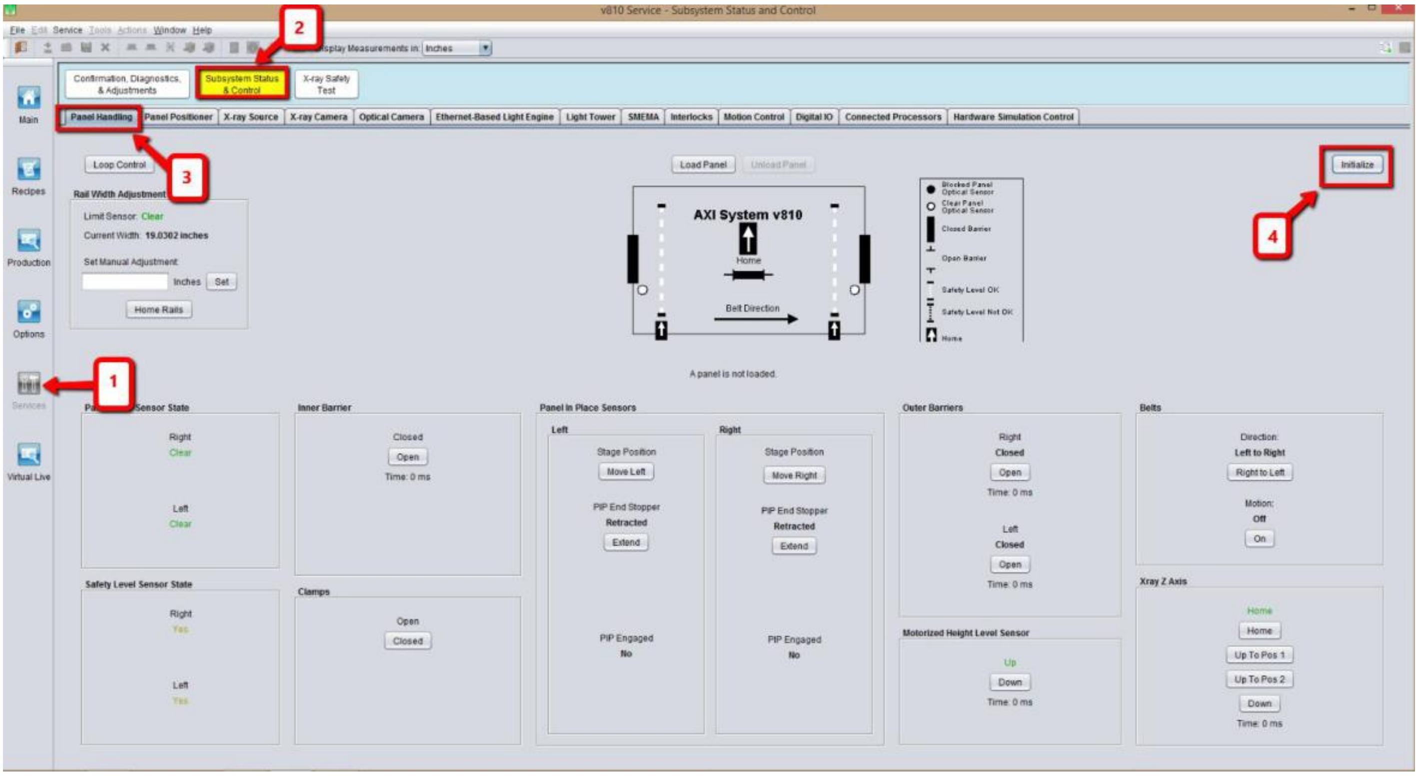


Figure 2 Panel Handling

4. Set the parameters as shown in the Figure 3 and click Load Panel.

   The outer barrier will open waiting to load the panel, click cancel

   Remarks : Panel width (inches) : 18

   Panel length (inches) : 18

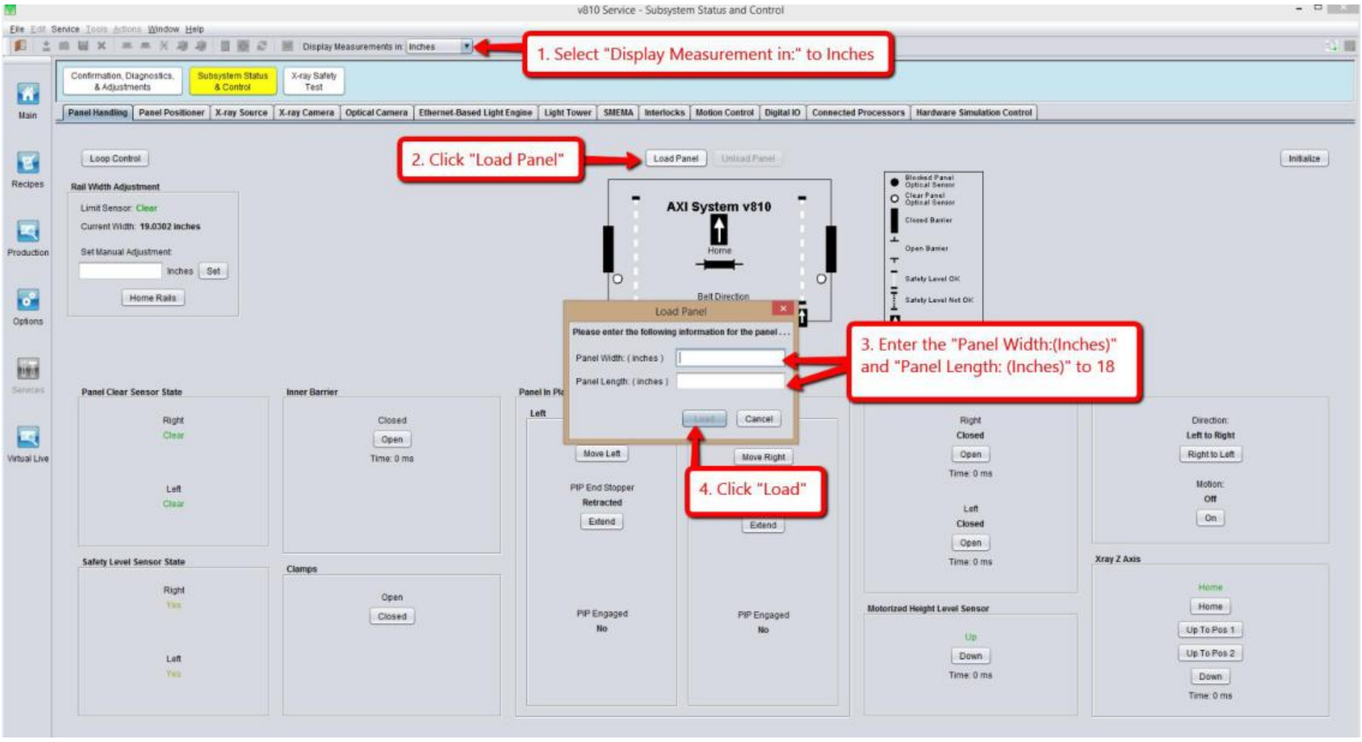


Figure 3 Panel Handling (Load Jig)

5. At Panel Handling tab, open left outer barrier
6. Close X-Ray Source & open front access door. Do not move the stage in both axis (X and Y axis require to remain in stationary position)
7. Power tune the Filter Height Sensor on the left outer barrier side by following Step (i) to Step (iii)

   i. Press the “Mode” button for more than 5 seconds as Figure 4

Figure 4 Filter Height Amplifier

ii. Progress bar of sensor amplifier will illustrate as Figure 5

Figure 5 Filter Height Power Tune Progress

iii. Filter height sensor intensity reading will be indicate within 1900~ 2200 as Figure 6

Figure 6 Filter Height Intensity Reading

8. Position 4XASS-357 with adjustment head (Figure 7) at Fixed Rail (Near to Left Outer Barrier) as Figure 8

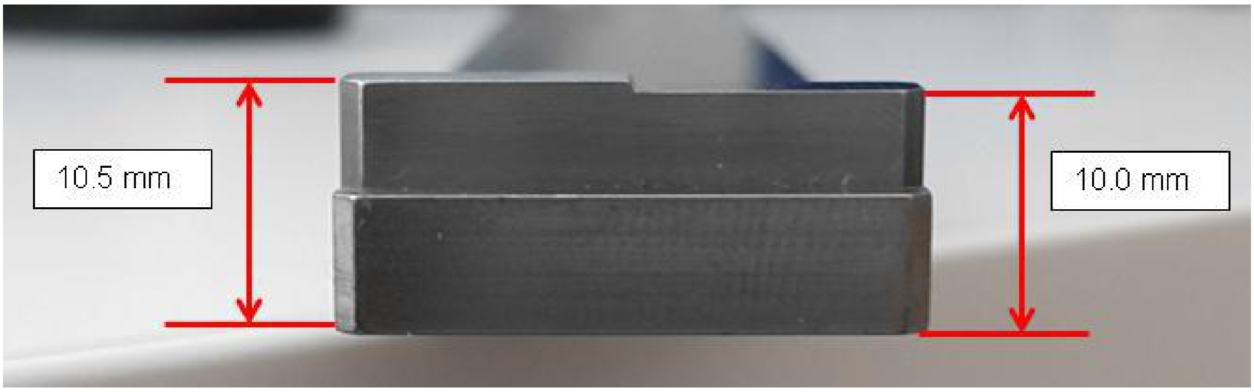


Figure 7 4XASS-357 Adjustment Head

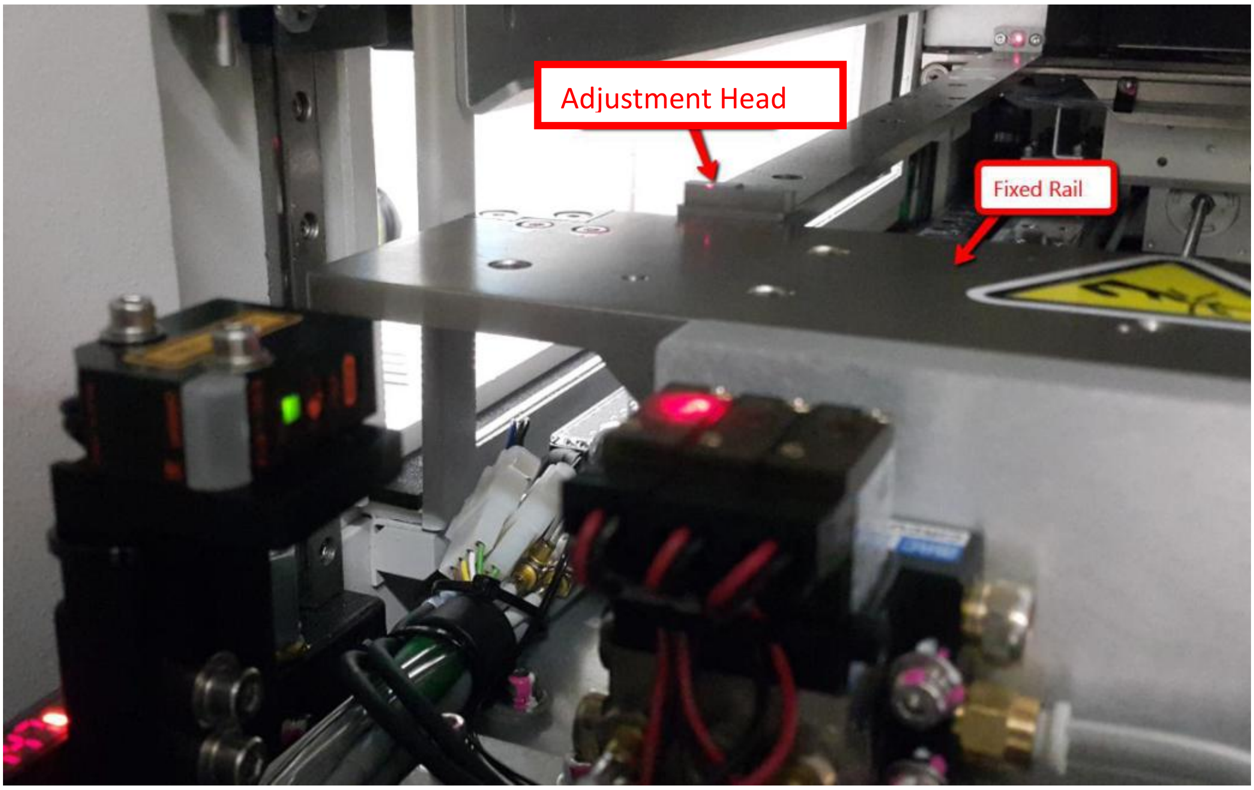


Figure 8 Filter Height Jig Position

9. Record filter height sensor intensity reading at 10.5 mm & 10.0 mm .May refer to Figure 9

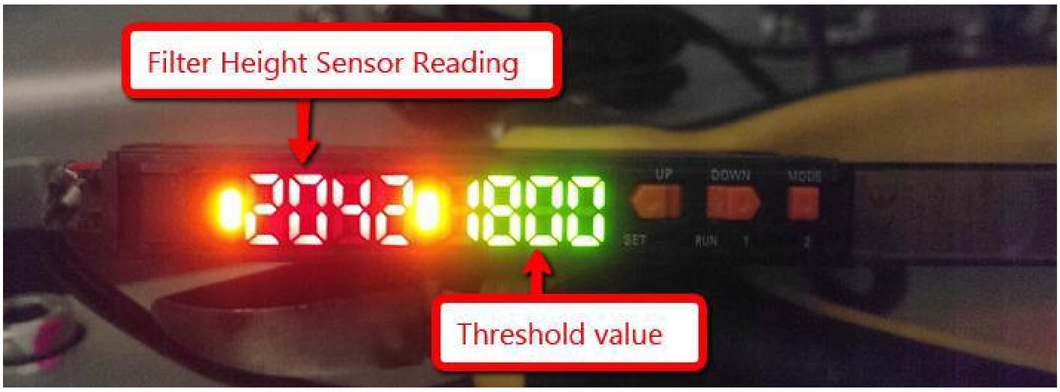


Figure 9 Filter Height Sensor

10. Next,rotate the adjustment head at Adjustable Rail as Figure 10 and record intensity reading for 10.5 mm & 10.0 mm

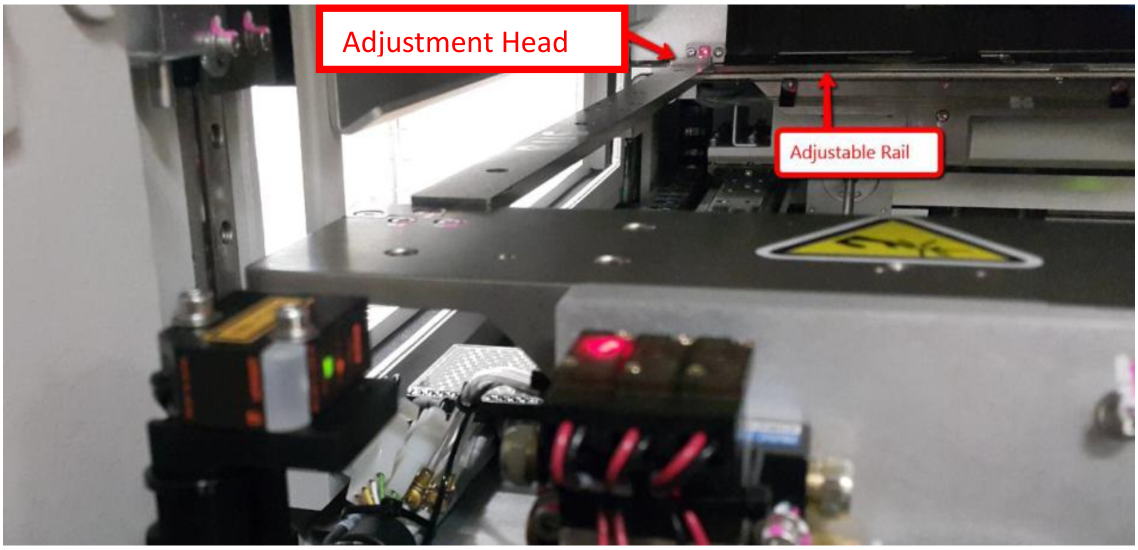


Figure 10 Filter Height Sensor Jig

11. Intensity reading which obtain from Step 8 ~ Step 10 require within expected spec as Table 1

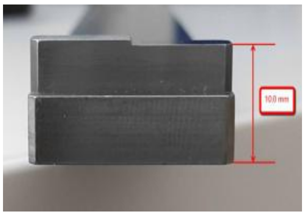
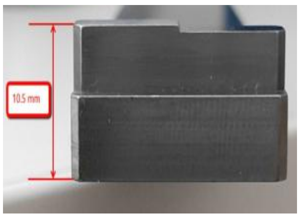
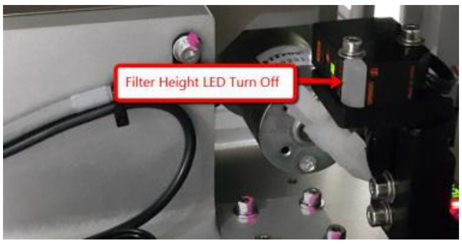
Jig Height	Sensor Amplifier Intensity Reading	Filter Height Sensor Status	DIO Input Bit
10.0 mm 	> 1850		
10.5 mm 	< 1750		

Table 1 Filter Height Checking

If actual intensity reading is not within expected spec Table 1, may proceed to Section B (Filter Height Adjustment)

B. Filter Height Adjustment Procedure

1. While XY Stage at Loading position,Loosen screws which mention in Figure 11 .Adjust filter sensor head until sensor spot is within middle of reflector as shown in the Figure 11.Tighten all screws

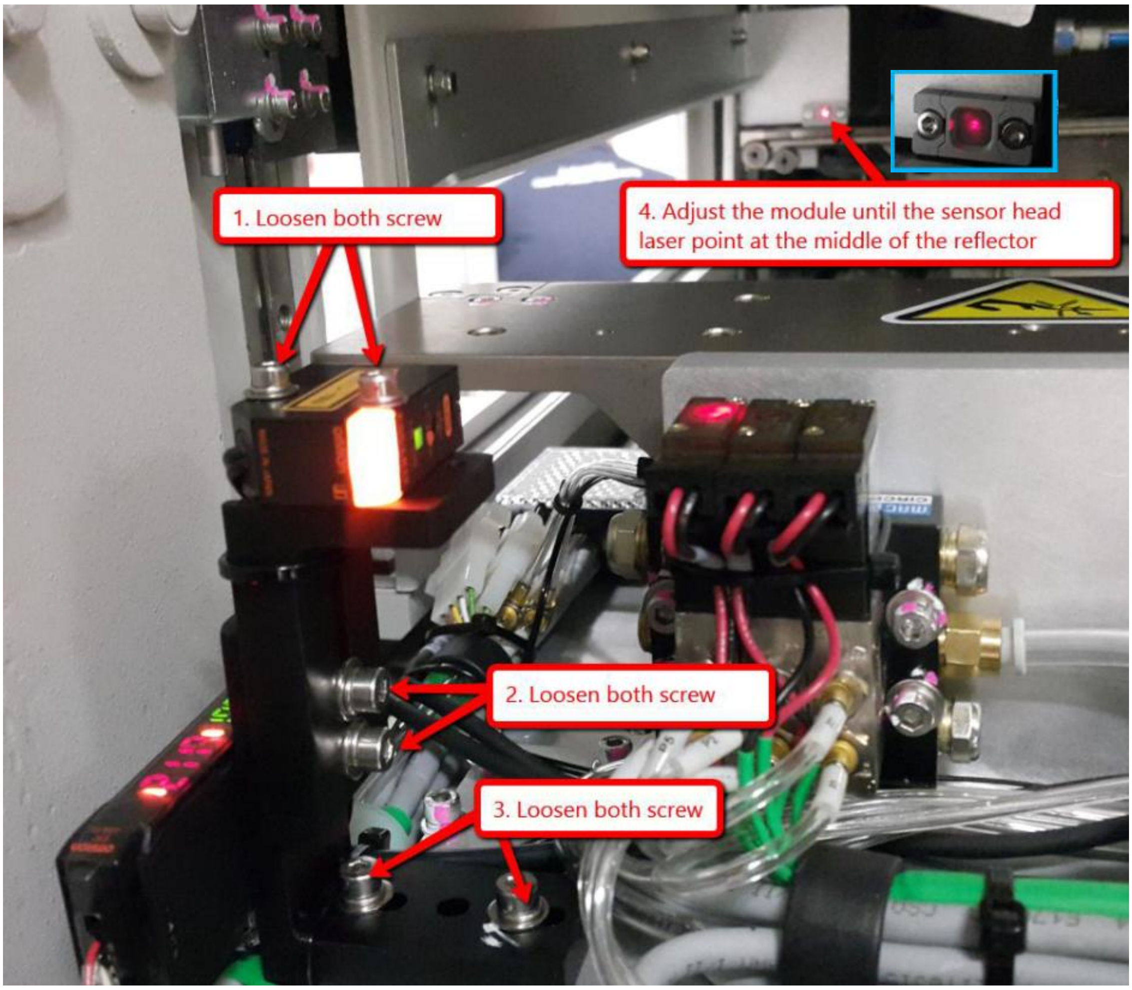


Figure 11 Filter Height Sensor Module Re-positioning

3. Position Filter Height Jig on AWA. Place M10 Allen Key respectively at front (Figure 13) and rear (Figure 14). May refer Step 3a ~ 3b to perform **simultaneous adjustment on** sensor height & angle where in order to meet expected result as Figure 13~Figure 14.

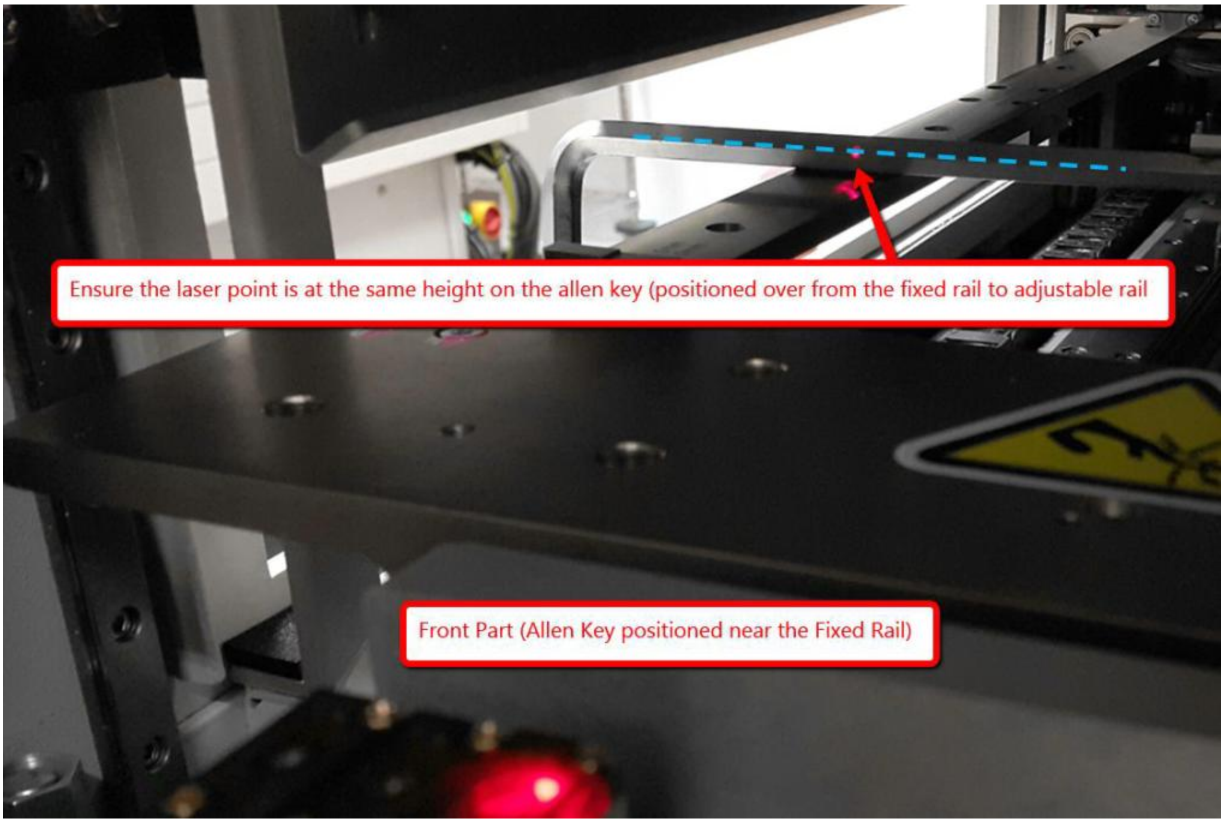


Figure 12 Filter Height Sensor Spot Linear Positioning

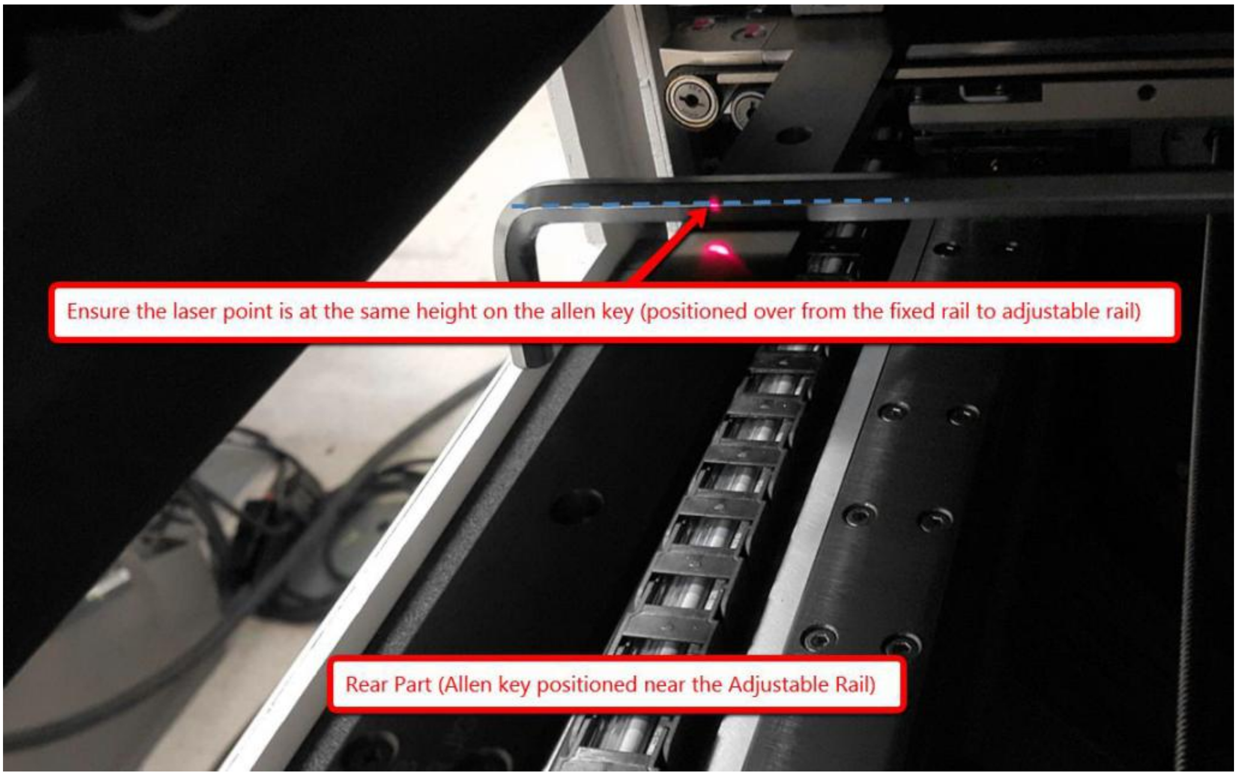


Figure 13 Filter Height Sensor Spot Linear Positioning

a) Loosen 2 screws to adjust the height of filter height module as Figure 14. Once the height is adjusted, tighten 2 screws.

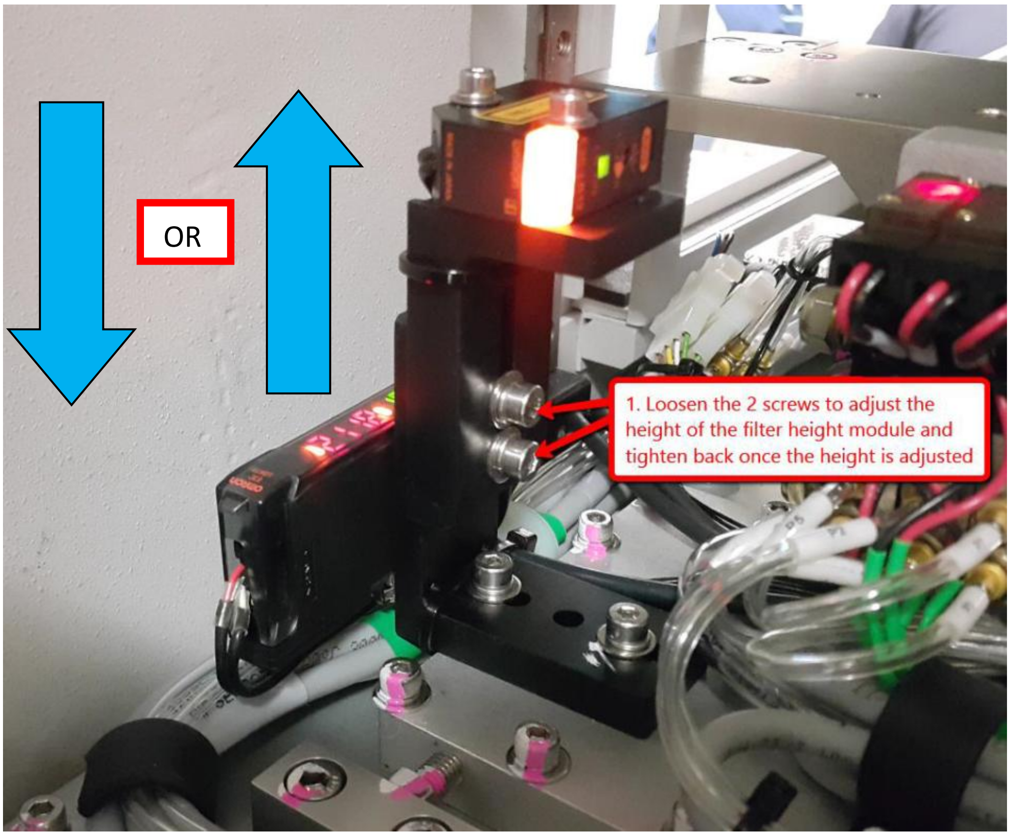


Figure 14 Filter Height Sensor Adjustment

b) Adjust the sensor angle by using the precision tool as Figure 15

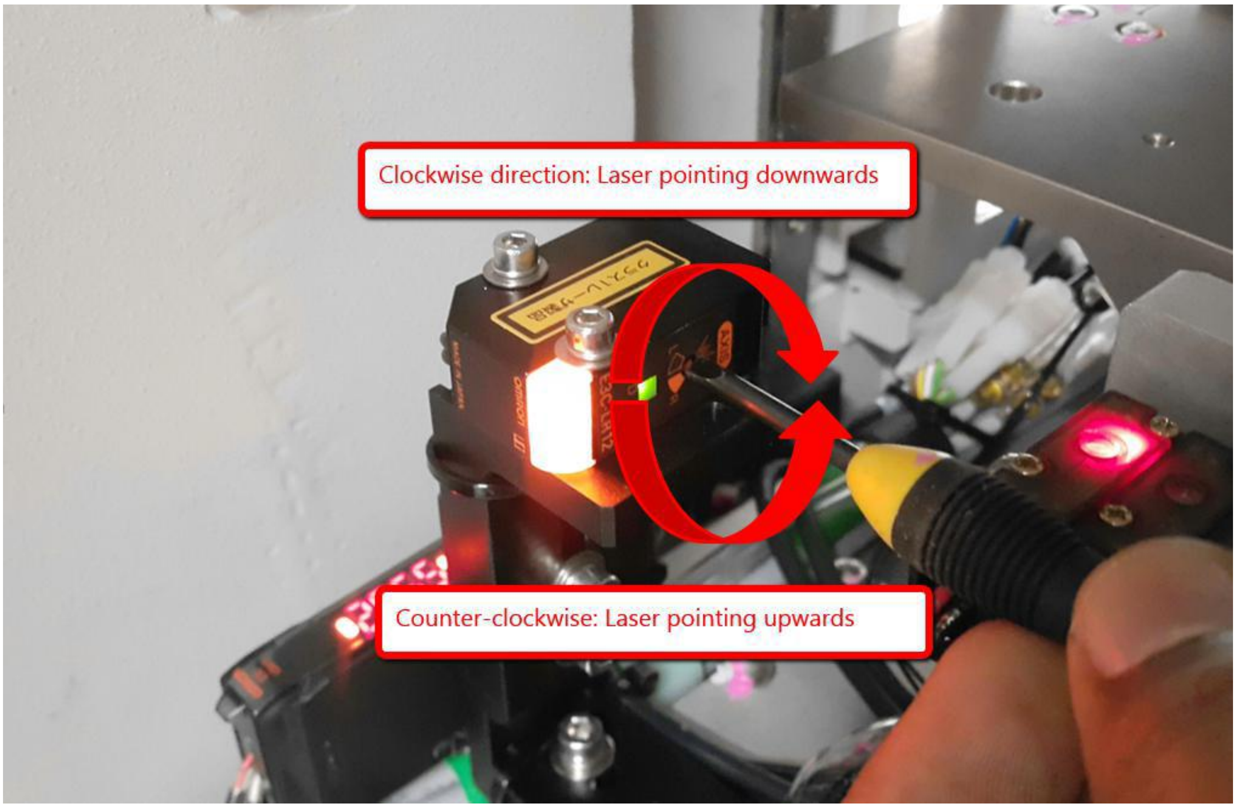


Figure 15 Filter Height Sensor Angle Adjustment

4. After Sensor Spot is within linear position, visual check sensor spot still within middle of reflector or any.If not, may require to perform minor adjustment on sensor head or height.

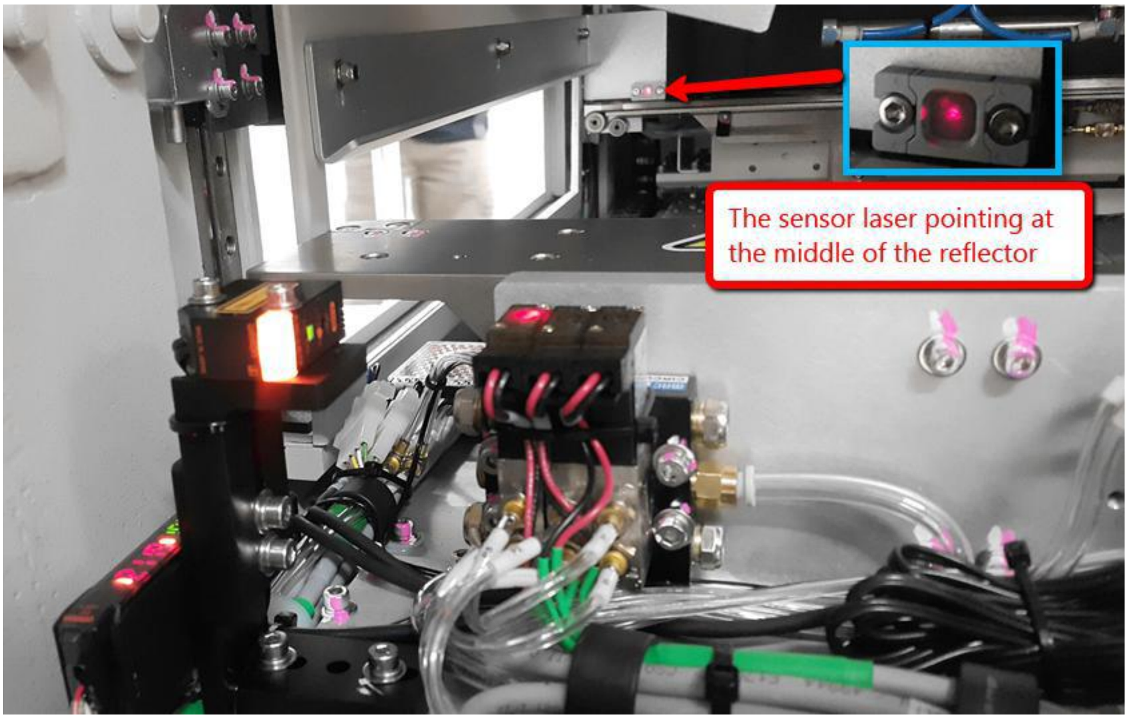


Figure 16 Sensor Spot Position

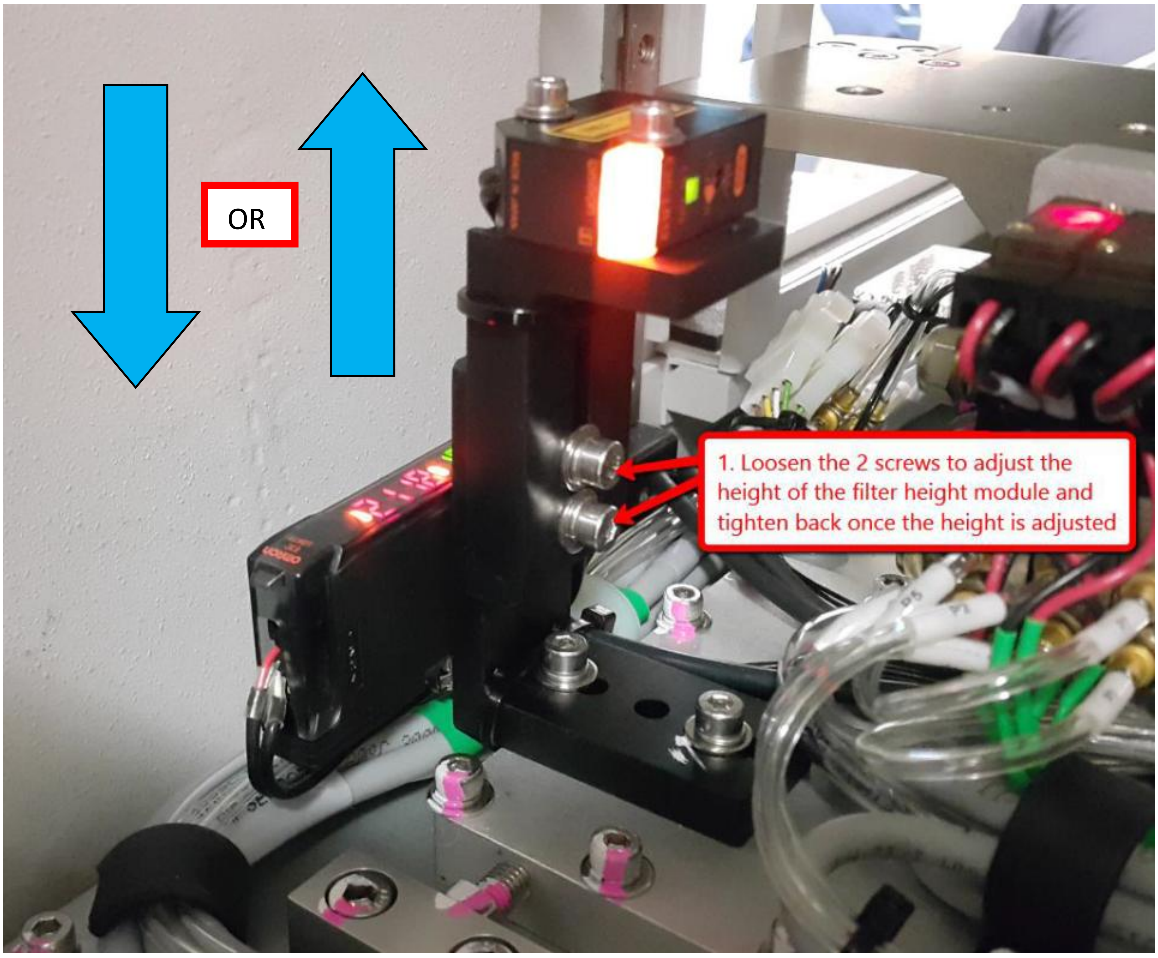


Figure 17 Sensor Height Adjustment

5. Power tune Filter Height Sensor by following Step (i) to Step (iii)

Remarks: Do not move the stage from loading position during amplifier power tuning.

i. Press the "Mode" button for more than 5 seconds as shown in the Figure 18

Figure 18 Filter Height Amplifier

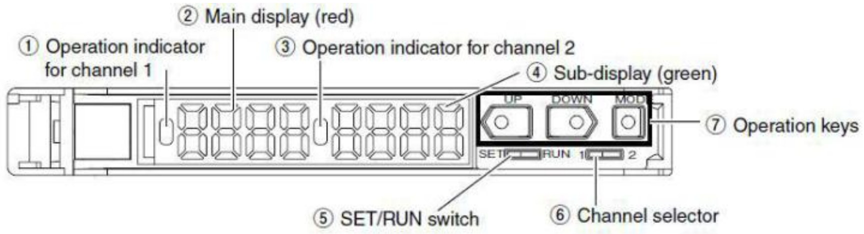
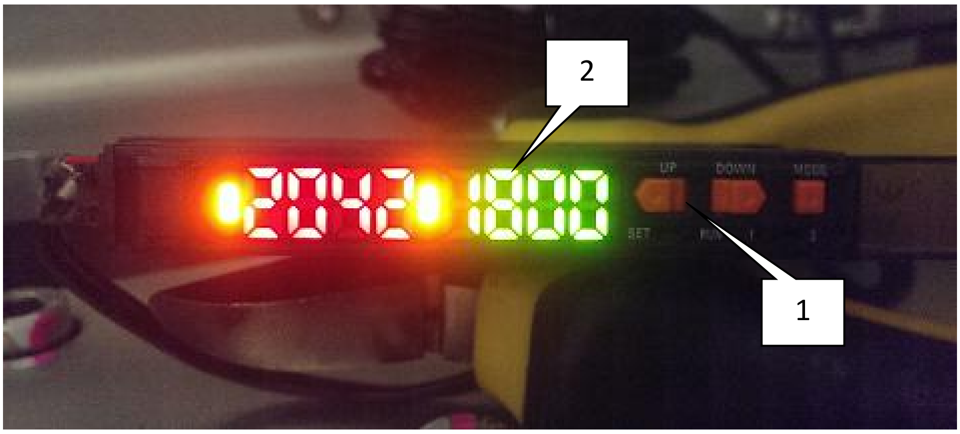
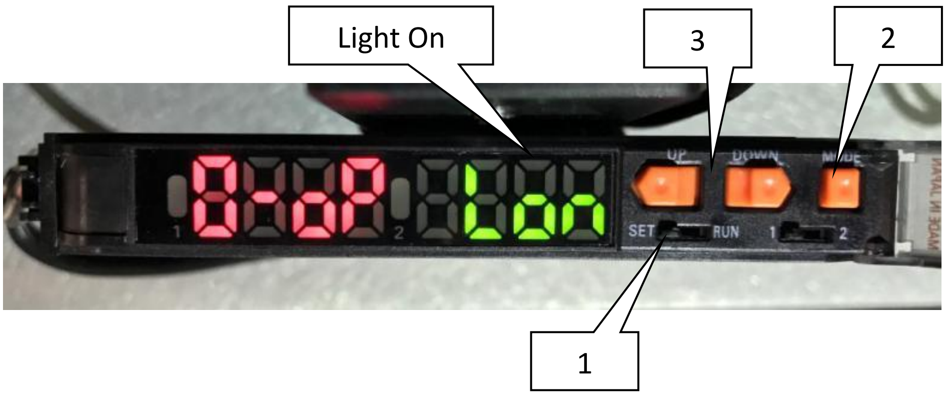
ii. The progress bar of the sensor amplifier will be as shown in the Figure 19

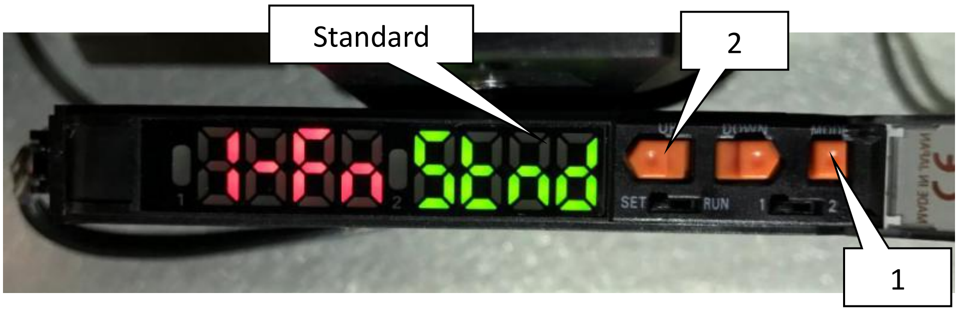
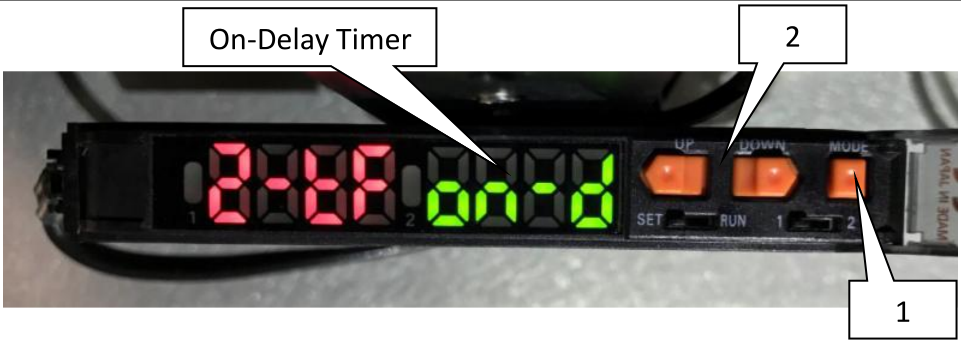
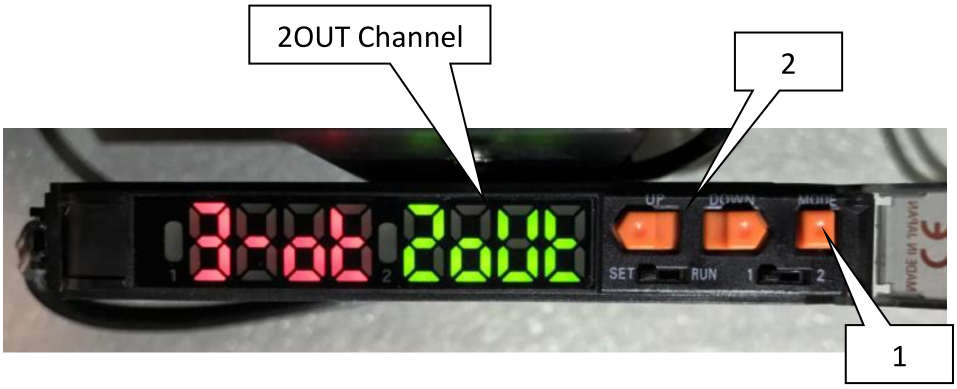
Figure 19 Filter Height Power Tune Progress

iii. The intensity reading of the filter height sensor laser will be approximately 1900 to 2200 as shown in the Figure 20

Figure 20 Filter Height Intensity Reading

8. Perform series of setting by refer to below table

No	Figure	Instruction
8.1		Remain stage at loading position. Open the Amplifier top cover, the button and display panel will show as attach.
8.2		After power tune the sensor, the intensity reading should around 2000 Perform quick calculation : Intensity reading -200 Example : 2042 -200 = 1842 Press the "UP/ DOWN" button and set the Green LED threshold value as "1842 "(Reference value Only).
8.3		1. Select SET for "SET/RUN switch". 2. Press "MODE" button to enter operation mode. 3. In operation mode press "UP/DOWN" button to select Light-On (Lon)

8.4		1. Press "MODE" button again to select detection method. In this stage select Standard (Stnd).
8.5		1. Press "MODE" button again to enter timer setting. 2. Press "UP/DOWN" to select On-Delay Timer (on-d)
8.6		1. Press "MODE" button again to enter timer time setting. 2. Press "DOWN" button until reach count 5000. # *default setting is 40 count, update the setting to 5000 count*
8.7		1. Press "MODE" button again to enter twin output setting. 2. Press "UP/ DOWN" button, select "2oUT" for output for each channel. *# It is due to only one channel output is used in our application and this setting allows us to isolate setting for channel two.*

8.9		1. Press “MODE” button again and ignore the other setting until the display back to the teaching interface. 2. Press the “UP/ DOWN” button and set the Green LED threshold value as 1800. 3. After complete the setting, turn the “SET/RUN switch” back to “RUN” position. 4. Power tune again the filter sensor amplifier and close the cover.

9. Place 4XASS-357 (Filter Height Jig) where adjustment head require next to Fixed Rail as shown in the Figure 21

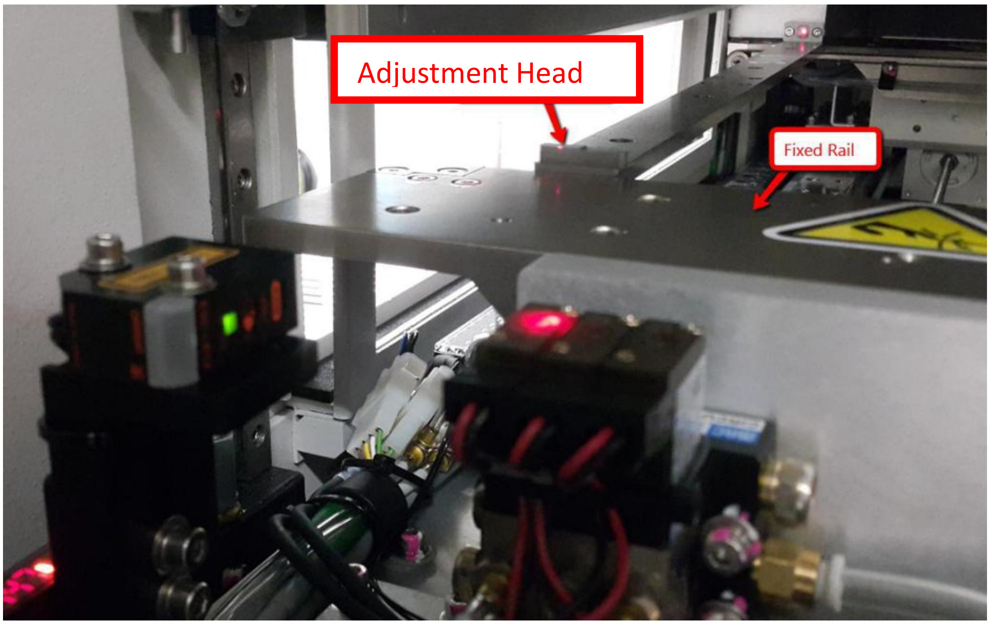


Figure 21 Filter Height Jig Position

10. By refer to expected spec in Table 2, perform sensor height adjustment as Figure 22

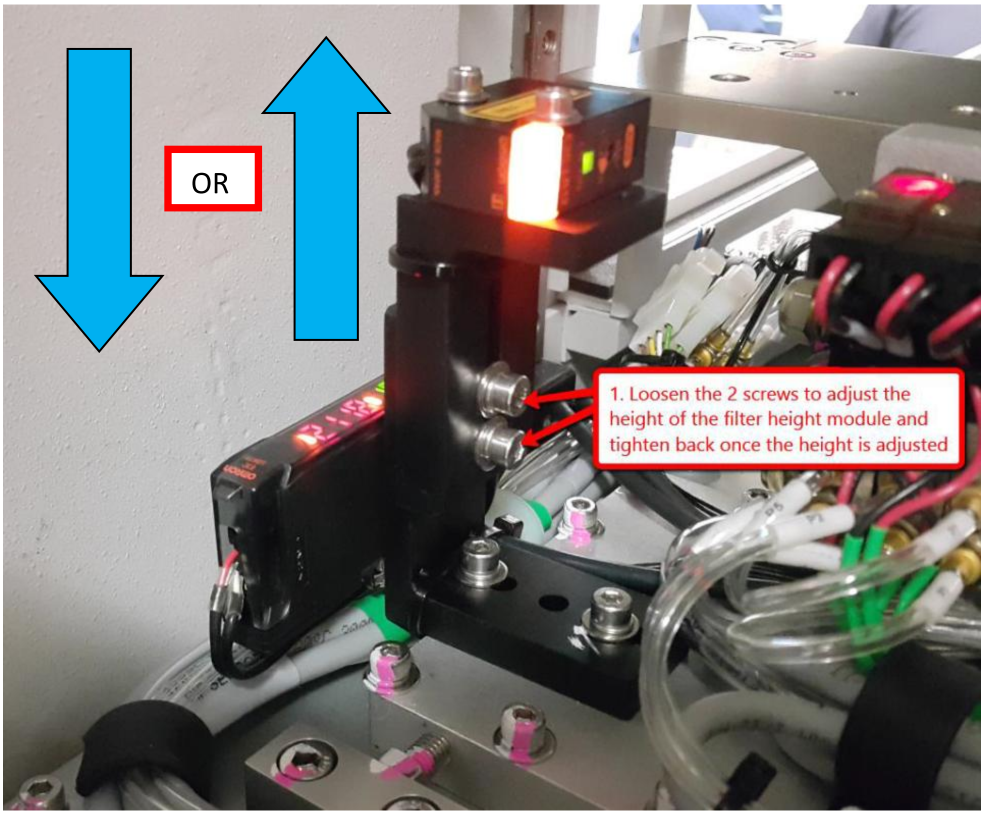


Figure 22 Filter Height Screw

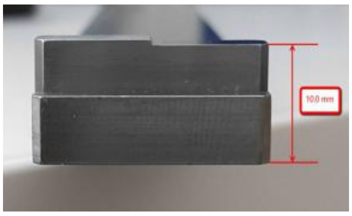
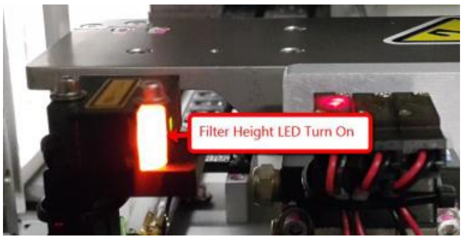
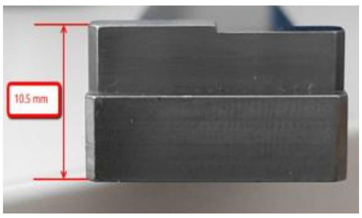
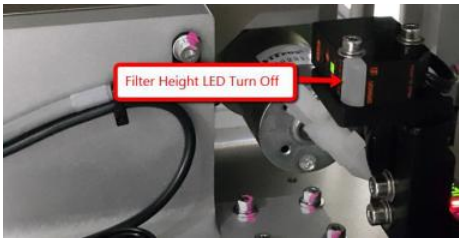
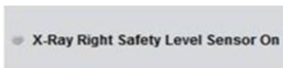
Jig Height	**Sensor Amplifier Reading**	**Filter Height Sensor Status**	**DIO Input Bit**
10.0 mm  	> 1850	 	
10.5mm  	< 1750	 	

Table 2 Filter Height Reading

11. Rotate the filter height jig where adjustment head require next to Adjustable Rail as Figure 23

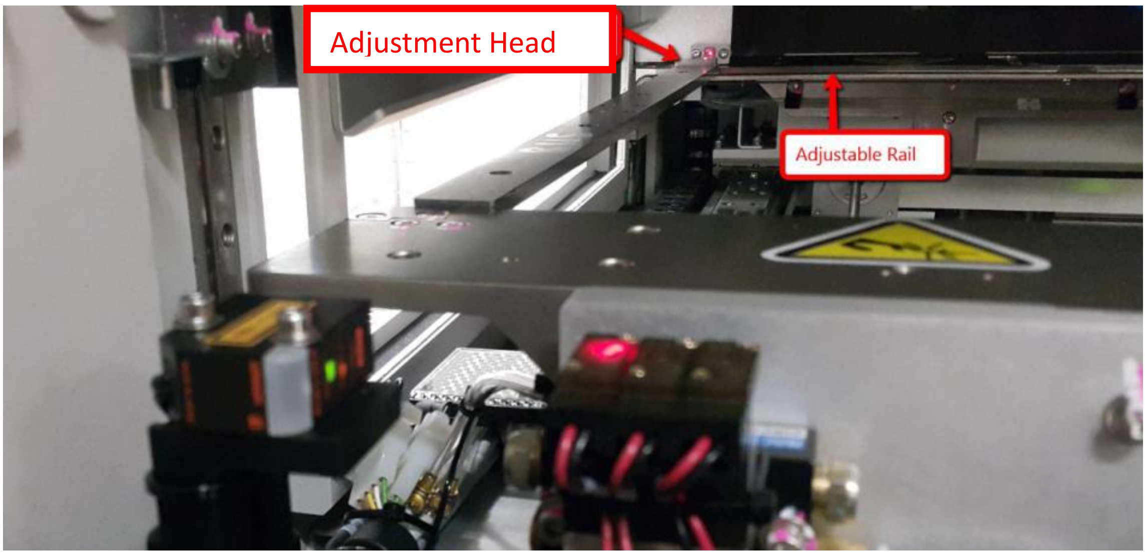


Figure 23 Adjustment Head

12. By refer to expected spec in Table 3, perform sensor height adjustment as Figure 24

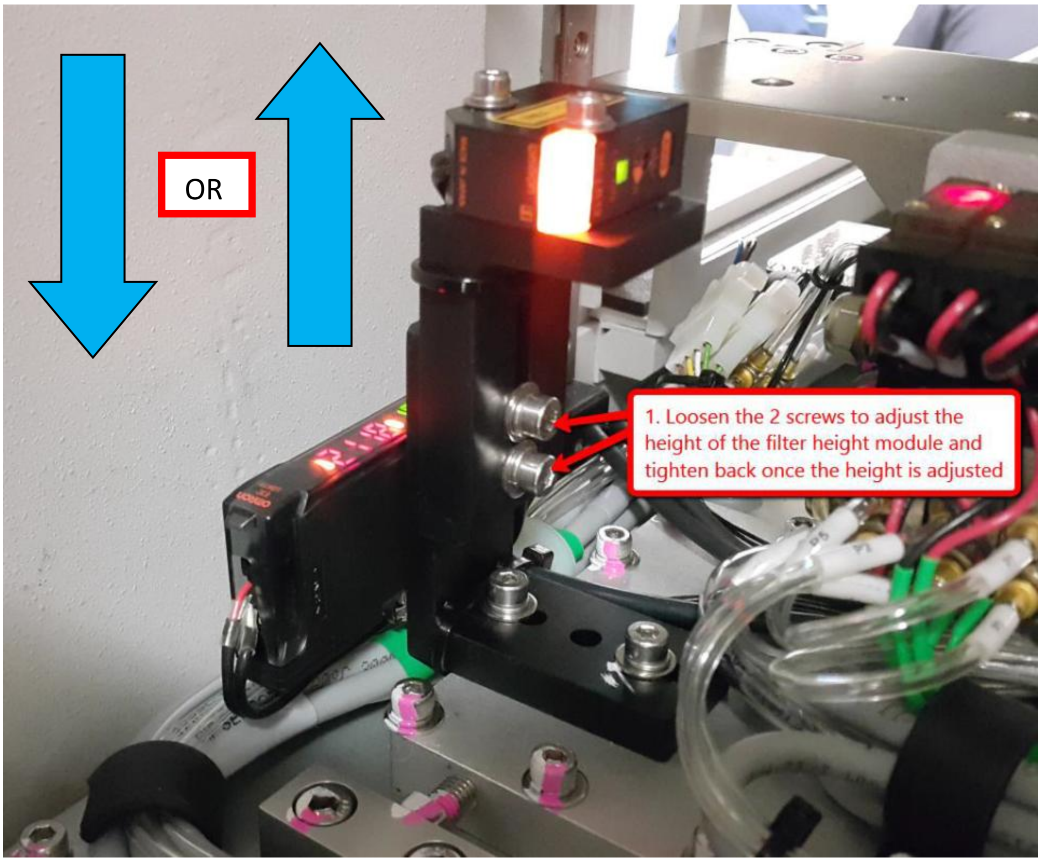


Figure 24 Filter Height Screw

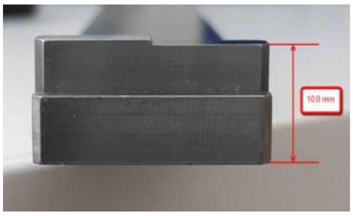
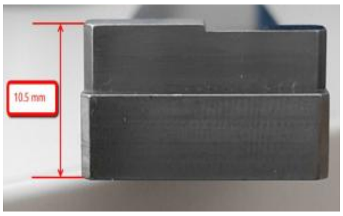
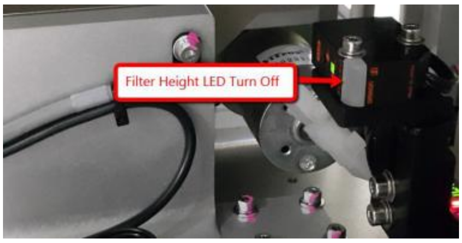
Jig Height	Sensor Amplifier Reading	Filter Height Sensor Status	DIO Input Bit
10.0 mm 	> 1850		
10.5mm 	< 1750		

Table 3 Filter Height Reading

13. Close the front access door left outer barrier.
14. Go to the Options -> Production Options -> General and set the Flow Direction as “Right To Left” as shown in the Figure 25

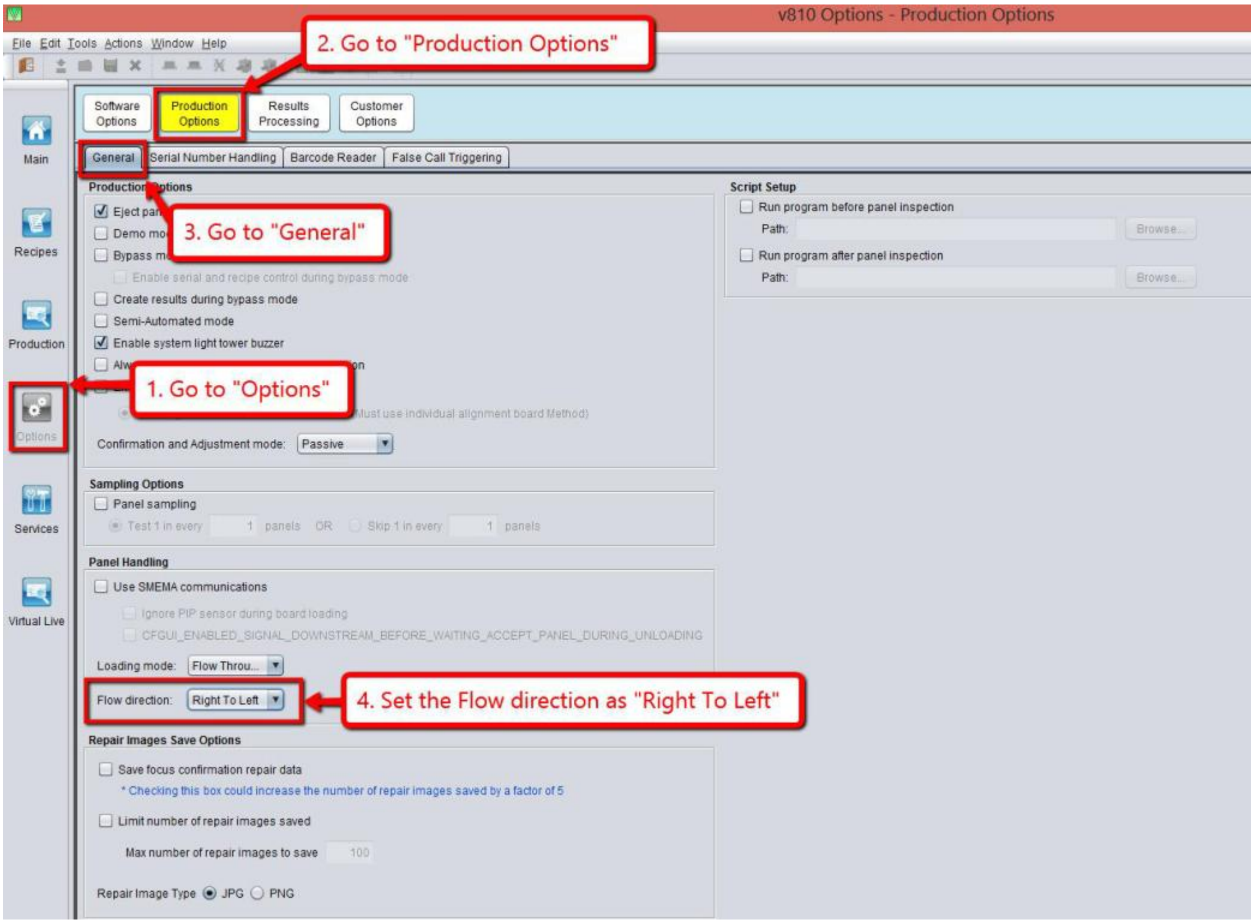


Figure 25 Loading mode setting

15. Repeat the steps in Section A for the verification of the Right Filter Height Sensor.